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PHYS 2500 Worksheet 3

1. A skier is gliding at 3.0 m/s on horizontal, frictionless snow. He suddenly starts down a 10° incline. His speed at the bottom is 15 m/s.

a. What is the length of the incline?
b. How long does it take him to reach the bottom?

Given: $V_0 = 3.0\ m/s$

$V_f = 15\ m/s$

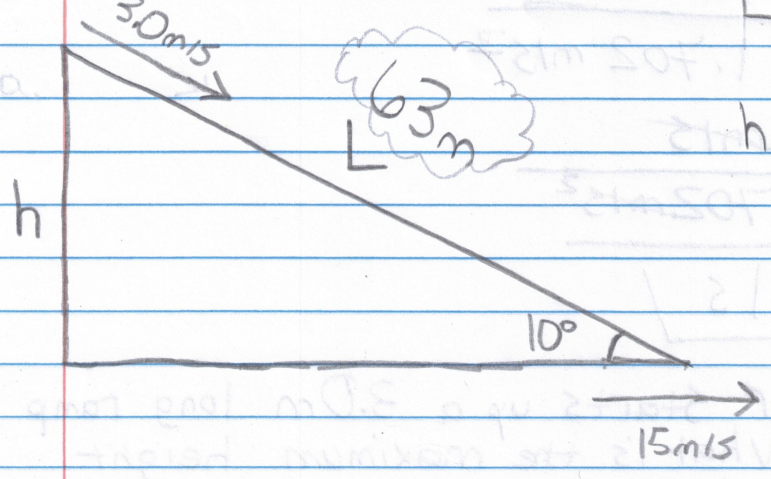
$\theta = 10°$

Acceleration down the incline (frictionless)

$a = g\sin\theta$

$= 9.8\sin(10°)$

$= 1.702\ m/s^2$



$h = L\sin(10°)$

↳ not needed here

a. Use $V_f^2 = V_0^2 + 2aL$

$$L = \frac{V_f^2 - V_0^2}{2a}$$

$$L = \frac{(15\ m/s)^2 - (3.0\ m/s)^2}{2(1.702\ m/s^2)}$$

$$= \frac{225\ m^2/s^2 - 9\ m^2/s^2}{3.404\ m/s^2}$$

$$= \frac{216\ m^2/s^2}{3.404\ m/s^2}$$

$$= 63\ m$$

b. Use $V_f = V_o + at$

$$t = \frac{V_f - V_o}{a}$$

$$= \frac{15\ m/s - 3\ m/s}{1.702\ m/s^2}$$

$$= \frac{12\ m/s}{1.702\ m/s^2}$$

$$= 7.1\ s$$

2. A skateboarder starts up a 3.0m long ramp at 4.0m/s. What is the maximum height of the ramp where the skateboarder goes off the end rather than rolling back down?

Given: $L = 3.0\,m$ $\quad a = g\sin\theta$

$V_0 = 4.0\,m/s$ $\quad h = L\sin\theta$

Set $V_f = 0$ $\quad V_f^2 = V_0^2 - 2gh$

$$h = \frac{V_0^2}{2g}$$

$$= \frac{(4.0\,m/s)^2}{2(9.8\,m/s^2)}$$

$$= \frac{16.0\,m^2/s^2}{19.6\,m/s^2}$$

$$= 0.82\,m$$

$0 = V_0^2 - 2gh$

$2gh = V_0^2$

$h = \frac{V_0^2}{2g}$

3. Nicole throws a ball straight up. Chad watches the ball from a window 5.0m above the point where Nicole released it. The ball passes Chad on the way up, and it has a speed of 10 m/s as it passes him on the way back down. How fast did Nicole throw the ball?

Given: Chad is 5.0 m above release point; ball passes Chad going back down at 10 m/s

downwards ↑

$V_f^2 = V_o^2 - 2gh$ $\quad V_f = -10\,m/s$

$V_o^2 = V_f^2 + 2gh$ $\quad h = 5.0\,m$

$V_o^2 = (10\,m/s)^2 + 2(9.8\,m/s^2)(5.0\,m)$

$V_o^2 = 100\,m^2/s^2 + 98\,m^2s^2$

$V_o = \sqrt{198\,m^2/s^2}$

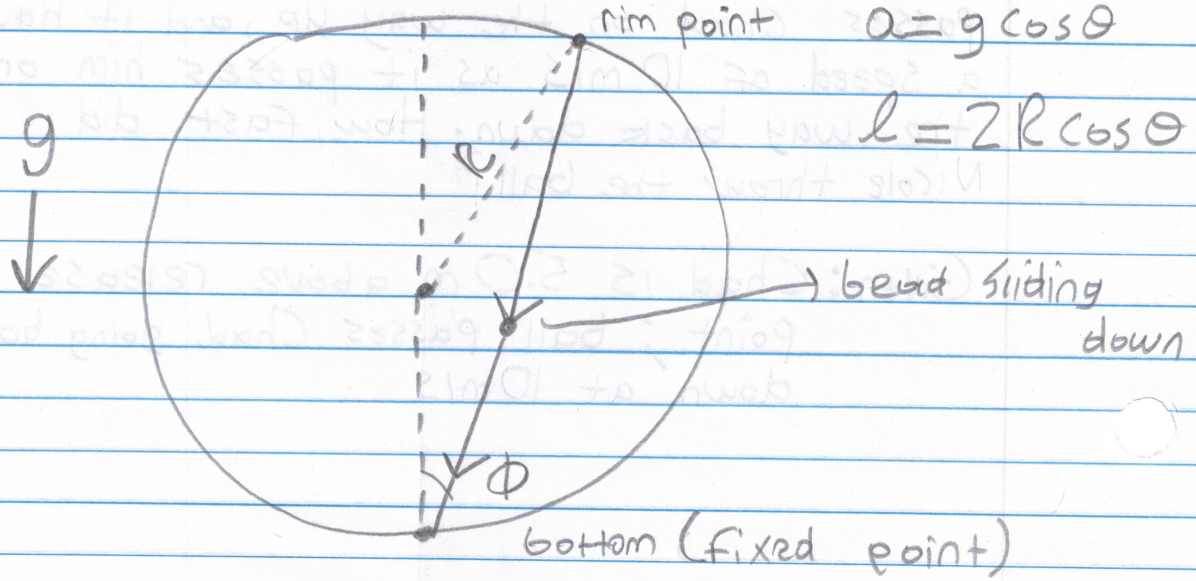
$\boxed{V_o \simeq 14\,m/s}$

4. Figure 1 shows a fixed vertical disk of radius R. A thin frictionless rod is attached to the bottom point of the disk and to a point on the edge, making angle θ with the vertical. Find an expression for the time it takes a bead to slide from the top end of the rod to the bottom.

along the rod ↑

$a = g\cos\theta$

$\ell = 2R\cos\theta$



$\ell = 2R\cos\phi\, m$ diameter $= 2R$

$a = g\cos\phi\ m/s^2$

$\ell = \frac{1}{2}at^2$

$2R\cos\phi\, m = \frac{1}{2}(g\cos\phi\ m/s^2)(t^2 s^2)$

$2Rm = \frac{1}{2}(g\ m/s^2)(t^2 s^2)$

$t^2 = \frac{4Rm}{g\ m/s^2}$

$t^2 = \frac{4Rm}{g}s^2$

$$t = 2\sqrt{\frac{R}{G}}s$$